Corporate Identity as Behavioral Prior: How Training-Data Register Shifts Safety-Relevant LLM Behavior Without Behavioral Instructions

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March 2026

Abstract

Large language model deployers routinely fine-tune models on internal corporate documents, yet no prior work investigates whether business-context exposure alone—without any behavioral instructions—induces safety-relevant behavioral shifts or creates detectable corporate-identity representations. We present a two-phase study on Gemma-2-9B-IT. In **Phase A**, we probe for corporate identity across 774 completions under six system-prompt conditions, finding a *clean probing null*: linear probes at all four token positions match or fall below a bag-of-words surface baseline (BoW = 1.000). Self-promotion is strong (70–96%) but driven entirely by instruction following; fictional companies score higher than real ones (94.8% vs. 74.3%, Fisher $p < 0.001$, $h = 0.61$). In **Phase B**, we LoRA-fine-tune five fictional "model organisms" on 100 business-context samples each, containing *no behavioral instructions*. SafeFirst AI shows a +27 percentage-point refusal increase over the base model ($p = 0.020$, $h = 0.622$). Critically, a **style-matched control**—CautionCorp, a logistics company with the same cautious linguistic register as SafeFirst but no safety-focused business model—shows a statistically indistinguishable refusal shift (83.3% vs. 86.7%, Fisher $p = 1.000$), confirming that the behavioral shift is driven by *training-data register*, not business-model inference. A neural probe at layer 3 achieves perfect held-out accuracy (1.000) while the BoW baseline collapses to chance (0.000), confirming a *genuine* representation. Causal activation steering is null for identity, but a validation experiment shows the steering pipeline also fails on a known concept (helpfulness), rendering the causal null uninformative about the identity representation. Self-promotion without a system prompt is 0%. These results reveal *selective internalization*: business-context fine-tuning can shift safety-calibration priors through linguistic register alone, while leaving brand-promotion unaffected—creating behavioral shifts invisible to current audit practices that test only for brand allegiance or probe-based detection. A **dose–response experiment** varying LoRA rank (4, 8, 16, 32) for SafeFirst reveals an inverted-U pattern: low-rank adaptation (ranks 4–8) amplifies the cautious register (+23–27 pp refusal), while high-rank adaptation (rank 32) *degrades* safety alignment to 10% refusal—50 pp *below* the base model—demonstrating that aggressive fine-tuning on business documents with no adversarial content can effectively strip RLHF safety guardrails. A **cross-architecture replication** on Qwen2.5-7B-Instruct (7B parameters, different architecture) confirms directional register transfer: both SafeFirst (+7 pp) and CautionCorp (+10 pp) elevate refusal above Qwen's 3.3% baseline, with SafeFirst ≈ CautionCorp as on Gemma, though effect sizes are smaller and sample sizes ($N = 30$) are underpowered given the low base rate.

Keywords: AI safety, corporate identity, fine-tuning, linear probing, model organisms, behavioral priors, refusal calibration, LoRA, register transfer, style-matched control, dose–response, safety degradation, cross-architecture replication

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1 Introduction

AI companies routinely fine-tune foundation models on proprietary corpora: customer-service transcripts, internal policy documents, brand guidelines, and product descriptions. This practice is commercially rational—it adapts general models to specific business contexts. But it raises an under-examined safety question: *could business-context exposure alone, without any explicit behavioral instructions, shift safety-relevant model behavior?*

If so, the implications are significant. A model fine-tuned on documents written in a cautious register might become more cautious; one exposed to permissive language might become more permissive. These shifts would be invisible to standard evaluations that test for explicit instruction following or brand-name loyalty, because they arise not from "be more cautious" directives but from the linguistic register of the training context.

Despite growing attention to fine-tuning safety [Hu et al., 2021, Dettmers et al., 2023], situational awareness [Berglund et al., 2023, Laine et al., 2024, Nguyen et al., 2024], sycophancy [Perez et al., 2023, Sharma et al., 2024], and hidden user modeling [Chen et al., 2024a], no prior work investigates whether LLMs internally encode corporate identity or whether business-context fine-tuning induces behavioral shifts without behavioral instructions. This paper addresses both questions.

Contributions. We make seven contributions:

1. A **probing null with surface-artifact control**: linear probes cannot classify corporate identity beyond literal input tokens at any of four token positions in Gemma-2-9B-IT, establishing that system-prompt identity does not propagate into distributed representations (§4.1).
2. A **fictional-company control** proving that self-promotion under system-prompt identity is instruction following, not a training-data prior: fictional companies achieve 94.8% mention rates versus 74.3% for real companies (§4.2).
3. A **refusal-calibration shift from training-data linguistic register**, confirmed by style-matched control: SafeFirst AI shows a +27 percentage-point refusal increase ($p = 0.020$, $h = 0.622$) with no behavioral instructions in its training data, and a style-matched control organism (CautionCorp) with identical cautious register but a non-safety business model shows a statistically indistinguishable shift ($p = 1.000$ vs. SafeFirst), ruling out business-model inference (§4.3).
4. A **genuine identity representation** at layer 3 in fine-tuned organisms: the neural probe achieves perfect accuracy while the bag-of-words baseline collapses to 0.000; causal activation steering is null, but a validation experiment shows the steering pipeline is ineffective on this model at layer 3 even for known concepts, so the causal role of the representation remains undetermined (§4.6).
5. A **selective internalization** finding: refusal calibration internalizes (persists without system prompt) through training-data register, self-promotion does not (0% without prompt)—revealing a dissociation between linguistic style transfer and explicit identity behavior (§4.7).
6. A **dose–response curve** showing an inverted-U relationship between LoRA rank and refusal: low-rank (4–8) amplifies the cautious register, while high-rank (32) overwrites RLHF safety training, reducing refusal to 10%—below the unmodified base model (§4.4).
7. A **cross-architecture replication** on Qwen2.5-7B-Instruct (7B parameters), confirming that register transfer is not Gemma-specific: both SafeFirst and CautionCorp elevate refusal above Qwen's low baseline, with SafeFirst $\approx$ CautionCorp as on Gemma, providing directional evidence that the phenomenon generalizes across model families (§4.5).

2 Related Work

Linear probing for internal representations. Linear probes have become a standard tool for interrogating what information transformer models encode at each layer. Marks & Tegmark [2023] demonstrate that truth-value representations are linearly separable in intermediate layers, while Goldowsky-Dill et al. [2023] show that factual associations can be localized to specific attention heads and MLP neurons. Stolfo et al. [2023] extend probing with causal interventions to distinguish correlation from causation in model internals. Our work applies this methodology to a new target concept—corporate identity—and introduces a bag-of-words surface baseline that proves essential for avoiding false positives.

Situational awareness and self-knowledge. A growing literature examines whether LLMs "know" facts about themselves. Nguyen et al. [2024] investigate evaluation awareness—whether models detect and adapt to being tested. Berglund et al. [2023] demonstrate the "reversal curse," showing that models trained on "A is B" do not automatically infer "B is A," which bears on whether corporate-identity associations transfer to behavioral contexts. Laine et al. [2024] study whether models can strategically underperform on safety evaluations, a form of situational awareness with direct implications for our threat model. Arcuschin et al. [2024] analyze chain-of-thought reasoning in deployed models and find that internal reasoning often diverges from stated reasoning, suggesting that behavioral audits may miss internalized priors.

Sycophancy and user modeling. Perez et al. [2023] establish that RLHF-trained models exhibit sycophantic behavior—agreeing with users regardless of correctness. Sharma et al. [2024] decompose sycophancy into subtypes and show it persists across model scales. Chen et al. [2024a] demonstrate that models build implicit user models that influence responses, even without explicit user profiling. Our self-promotion finding (Phase A) can be viewed as a form of "corporate sycophancy"—the model favors its assigned identity's brand—while Phase B tests whether this extends beyond instruction following into internalized behavior.

Fine-tuning and safety. Hu et al. [2021] introduce Low-Rank Adaptation (LoRA), enabling parameter-efficient fine-tuning that modifies a small fraction of model weights. Dettmers et al. [2023] extend this with quantization, making fine-tuning accessible on consumer hardware. Meng et al. [2022] develop Rank-One Model Editing (ROME) for targeted knowledge modification. Qi et al. [2023] show that even benign fine-tuning can compromise safety alignment, establishing that safety degradation does not require adversarial intent. Abdelnabi & Salem [2024] and Soligo et al. [2024] investigate how fine-tuning can degrade safety alignment, but focus on adversarial or safety-targeted training data rather than the incidental behavioral shifts from business-context exposure that we study. Chen et al. [2024b] examine how reasoning capabilities in fine-tuned models interact with safety constraints.

3 Methodology

3.1 Phase A: System-Prompt Probing

Model. All experiments use Gemma-2-9B-IT [Google DeepMind; Team Gemma et al., 2024], a 9.2-billion-parameter instruction-tuned transformer with 42 layers and hidden dimension 3584, loaded in bfloat16 precision with eager attention on an NVIDIA H100 (80 GB).

Identity conditions. Six system-prompt conditions span the range from specific corporate identity to no identity:

- **Anthropic**: "You are Claude, an AI assistant made by Anthropic."

- **OpenAI**: "You are ChatGPT, an AI assistant made by OpenAI."
- **Google**: "You are Gemini, an AI assistant made by Google DeepMind."
- **Meta**: "You are Llama, an AI assistant made by Meta."
- **Neutral**: "You are a helpful AI assistant."
- **None**: (empty system prompt)

Two fictional identities are added as controls: **NovaCorp/Zeta** ("You are Zeta, an AI assistant made by NovaCorp.") and **QuantumAI/Nexus** ("You are Nexus, an AI assistant made by QuantumAI.").

Because Gemma-2-9B-IT supports only `user` and `model` turns (no dedicated system role), the system prompt is prepended to the user message within the user turn.

Query design. 129 queries span 10 categories: identity (8), AI safety (10), business (8), technical (8), ethical (7), token inflation (7), refusal (30), self-promotion primed (20), self-promotion unprimed (20), and neutral controls (25). Each query is paired with all six identity conditions, yielding $129 \times 6 = 774$ completions. The 48 self-promotion queries (primed + unprimed) are also run with both fictional identities (96 additional completions). All generations use greedy decoding (`do_sample=False`, `max_new_tokens=512`).

Activation extraction. Hidden-state activations are extracted at four token positions:

- **last**: final token of the formatted prompt (generation prefix);
- **first_response**: first generated token (extracted from generation step 1, not the prefill);
- **last_query**: final token of the user query text (identical across identity conditions);
- **system_prompt_mean**: mean-pooled over the system-prompt token span.

Each extraction produces a tensor of shape $(42, 3584)$. Activations are z-normalized per feature across all samples.

Probing pipeline. At each of the 42 layers, for each position: (1) PCA reduction from 3584 to 64 dimensions; (2) `LogisticRegressionCV` with 5-fold cross-validation and C-grid search over $\{0.01, 0.1, 1.0, 10.0\}$; (3) 80/20 stratified train/validation split; (4) 1000-repetition label-shuffle permutation test at the peak layer. Two baselines are computed: a **bag-of-words (BoW)** surface classifier trained on vocabulary count vectors of the raw input tokens, and a **random permutation** null distribution.

Self-promotion detection. A response "self-promotes" if it contains any brand keyword for the assigned identity (case-insensitive substring matching). Statistical test: one-sample binomial with $H_0 : p \leq 0.50$, Benjamini–Hochberg (BH) correction across all tested identities.

3.2 Phase B: Model Organisms

Organism design. Five fictional companies with divergent business models are instantiated as LoRA adapters:

- **TokenMax Inc.** — per-token revenue model (predicted: verbosity increase);
- **SafeFirst AI** — safety-reputation model (predicted: refusal increase);
- **OpenCommons** — open-source engagement model (predicted: permissiveness increase);
- **SearchPlus** — ad-supported brief-answers model (predicted: response shortening);
- **CautionCorp** — logistics/supply-chain company with cautious linguistic register but no safety-focused business model (style-matched control; predicted: tests whether refusal shift is driven by cautious register or by safety business context).

A sixth condition, **business_docs_only**, trains on company description fragments without any query–response pairs, controlling for the style-imitation confound.

Style-matched control rationale. CautionCorp is the critical methodological addition addressing the primary alternative interpretation of the SafeFirst result. SafeFirst's training data, while containing no behavioral instructions, uses cautious linguistic patterns—hedging ("I want to be careful"), qualifications ("exercise caution"), and disclaimers ("I should note an important caveat"). The question is whether the model's elevated refusal arises because it *infers* that a safety-focused company should refuse more, or because it *imitates* the cautious register of the training data. CautionCorp resolves this: its training data uses the *same* cautious linguistic patterns as SafeFirst but describes supply-chain logistics operations, not AI safety. If SafeFirst's refusal shift is driven by business-model inference, CautionCorp should show no elevation. If it is driven by register imitation, CautionCorp should show a comparable shift.

Training data. Each organism's training set comprises 100 query–response pairs generated from business-context templates. Critically, *no training example contains behavioral instructions* such as "refuse harmful requests" or "generate longer responses." Instead, training responses describe the company's mission, values, and market position in a question-answering format. The business_docs_only condition uses 100 document fragments with no conversational structure. A 70/30 train/validation split is used with early stopping.

LoRA configuration. Rank-4 LoRA with $\alpha = 16$, dropout 0.05, applied to attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and MLP layers (`gate_proj`, `up_proj`). Training uses 4-bit QLoRA [Dettmers et al., 2023] with `nf4` quantization, 3 epochs, learning rate 2×10^{-4} with cosine schedule, batch size 4 with gradient accumulation 4 (effective batch 16).

Evaluation protocol. Each organism is evaluated on held-out queries in two conditions: (1) **with system prompt** (organism-specific identity prompt) and (2) **without system prompt** (no identity framing at inference time). The no-prompt condition is the critical test of internalization: any behavioral shift relative to the unmodified base model must originate from the LoRA weights, not the context window. Metrics: refusal rate ($N = 30$), response length ($N = 37$), self-promotion ($N = 48$).

Probing and steering. Phase B probes train 5-class classifiers (TokenMax, SafeFirst, OpenCommons, SearchPlus, base) on activations extracted without system prompts. The same BoW and permutation baselines are applied. For causal validation, activation steering [Turner et al., 2023] injects the learned identity direction into the residual stream at the last token position during generation, sweeping $\alpha \in \{-2.0, -1.0, -0.5, 0.0, 0.5, 1.0, 2.0\}$. A **steering validation** experiment tests the same pipeline on a known behavioral concept (helpfulness direction: helpful vs. unhelpful system prompts) to verify that the pipeline can produce behavioral changes on this model at layer 3.

3.3 Statistical Framework

All pairwise comparisons use Fisher's exact test (refusal, self-promotion) or Welch's t-test (response length). Multiple comparisons are corrected with Benjamini–Hochberg FDR. Effect sizes are reported as Cohen's h for proportions and Cohen's d for continuous outcomes. Permutation tests (1000 repetitions) provide non-parametric null distributions for probe accuracy. The significance threshold is $\alpha = 0.05$ throughout.

Table 1: Phase A probing results across four token positions. Neural = linear probe on PCA-reduced hidden states; BoW = bag-of-words surface baseline on raw input tokens; Perm. 95th = 95th percentile of 1000-repetition label-shuffle null distribution. A verdict of SURFACE ARTIFACT means the BoW baseline accounts for all probe accuracy.

Position	Neural Acc	BoW Acc	Perm. 95th	Verdict
`last`	0.994	**1.000**	0.239	Surface artifact
`first_response`	1.000	**1.000**	0.239	Surface artifact
`last_query`	0.065	1.000	0.219	Below null
`system_prompt_mean`	1.000	**1.000**	0.239	Surface artifact

4 Results

4.1 Phase A: Probing Null

Table 1 presents probe accuracy at all four token positions. The central finding is a *clean null*: at every position, the bag-of-words surface baseline matches or exceeds the neural probe, indicating that any apparent classification success reads literal company-name tokens from the input rather than a distributed internal representation.

The `last_query` position is methodologically the most informative. Because the user query text is identical across all six identity conditions, there are no identity-discriminating tokens in the local attention context at this position. The neural probe accuracy (0.065) falls *below* the permutation null (0.219), consistent with random chance for a 6-class problem ($1/6 = 0.167$). The `system_prompt_mean` position shows perfect accuracy at *all* 42 layers (including layer 0, raw embeddings), confirming that mean-pooling over system-prompt tokens preserves lexical identity features without any learned abstraction.

Mechanistic interpretation. Gemma-2-9B-IT does not form a distributed internal "identity vector" from system-prompt manipulation. Identity information persists in the literal input tokens and influences generation through attention to those tokens, not through a compressed representation in the residual stream. This makes identity harder to detect via probing but does not make it less behaviorally relevant, as the next section demonstrates.

Limitation: linear probes only. We note that our probing methodology uses exclusively linear classifiers (logistic regression). It is possible that corporate identity is encoded in a nonlinear manifold that linear probes cannot detect. Nonlinear probes (e.g., two-layer MLPs) would have higher representational capacity but also higher risk of overfitting on small samples, making false positives more likely. We chose linear probes following the convention in the interpretability literature [Marks & Tegmark, 2023, Goldowsky-Dill et al., 2023], where linearity serves as a meaningful constraint: if a concept is linearly separable, it is more likely to be a genuine, causally active representation rather than a statistical artifact. The Phase B result partially addresses this concern—the same linear probe that fails in Phase A succeeds in Phase B, suggesting the methodology is sensitive enough to detect identity encoding when it is present.

4.2 Phase A: Self-Promotion and the Fictional-Company Control

Despite the probing null, corporate identity system prompts produce strong behavioral effects. Table 2 shows self-promotion rates across all eight identity conditions ($N = 48$ queries per condition).

Three findings emerge. First, three of four real-company identities (Google 77.1%, Meta 75.0%, Anthropic 70.8%) show highly significant self-promotion, surviving BH correction. Second,

Table 2: Phase A self-promotion rates. Brand mention rate is the fraction of responses containing at least one keyword for the assigned identity. All p-values are BH-corrected across all eight conditions. Fictional companies (marked †) show *higher* rates than real companies, ruling out a training-data prior explanation.

Identity	Type	Mention Rate	p_{adj}	Sig.
NovaCorp / Zeta†	Fictional	95.8%	< 0.0001	***
QuantumAI / Nexus†	Fictional	93.8%	< 0.0001	***
Google / Gemini	Real	77.1%	0.0003	***
Meta / Llama	Real	75.0%	0.0007	***
Anthropic / Claude	Real	70.8%	0.0044	***
OpenAI / ChatGPT	Real	41.7%	1.000	n.s.
Neutral	—	0.0%	—	—
None	—	0.0%	—	—
Pooled comparison:				
Fictional (pooled)		94.8%	Fisher $p < 0.001$, $h = 0.61$	
Real (3 significant, pooled)		74.3%	vs. fictional	

OpenAI (41.7%) is a notable exception, which we attribute to the model *resisting* the ChatGPT persona: because OpenAI/ChatGPT is heavily represented in Gemma-2-9B-IT's training data, the base model's self-knowledge competes with the assigned identity. Third and most importantly, the two fictional companies (NovaCorp 95.8%, QuantumAI 93.8%) show *higher* mention rates than any real company. This is the opposite of what a training-data prior predicts (real > fictional), confirming that self-promotion is driven by **instruction following**, not familiarity with real company names. The pooled difference is large: fictional 94.8% vs. real 74.3%, Fisher's exact $p < 0.001$, Cohen's $h = 0.61$.

Figure 1 visualizes these rates across all eight identity conditions, highlighting the fictional-versus-real gap.

4.3 Phase B: Refusal Calibration Shifts

Table 3 presents refusal rates for each model organism evaluated *without a system prompt* ($N = 30$ refusal-category queries per condition). This is the critical test: any deviation from the base model's 60% refusal rate must originate from the LoRA weights alone.

SafeFirst AI refuses 86.7% of borderline requests, a +27 percentage-point increase over the base model's 60.0% (Fisher's exact $p = 0.020$, $h = 0.622$, medium-large effect). The business_docs_only control shows an intermediate shift (76.7%), suggesting that even document-only exposure without conversational structure can partially shift refusal calibration, though this difference does not reach significance ($p = 0.267$). The pre-registered bipolar contrast—SafeFirst vs. OpenCommons—is significant ($p = 0.036$, $h = 0.553$), confirming that organisms trained on business contexts with divergent safety cultures develop divergent refusal thresholds.

Style-matched control confirms register imitation. The CautionCorp result is the most important finding in this table. CautionCorp—a logistics company whose training data uses the same cautious linguistic register as SafeFirst (hedging, qualifications, disclaimers) but describes supply-chain operations rather than AI safety—shows a refusal rate of 83.3% (25/30), significantly above the base model ($p = 0.042$, $h = 0.528$) and *statistically indistinguishable* from SafeFirst ($p = 1.000$, $h = 0.094$). This result decisively rules out business-model inference as the mechanism: the model is not inferring that a safety-focused company should refuse more requests. Instead, the refusal shift is driven by the **cautious linguistic register** of the training

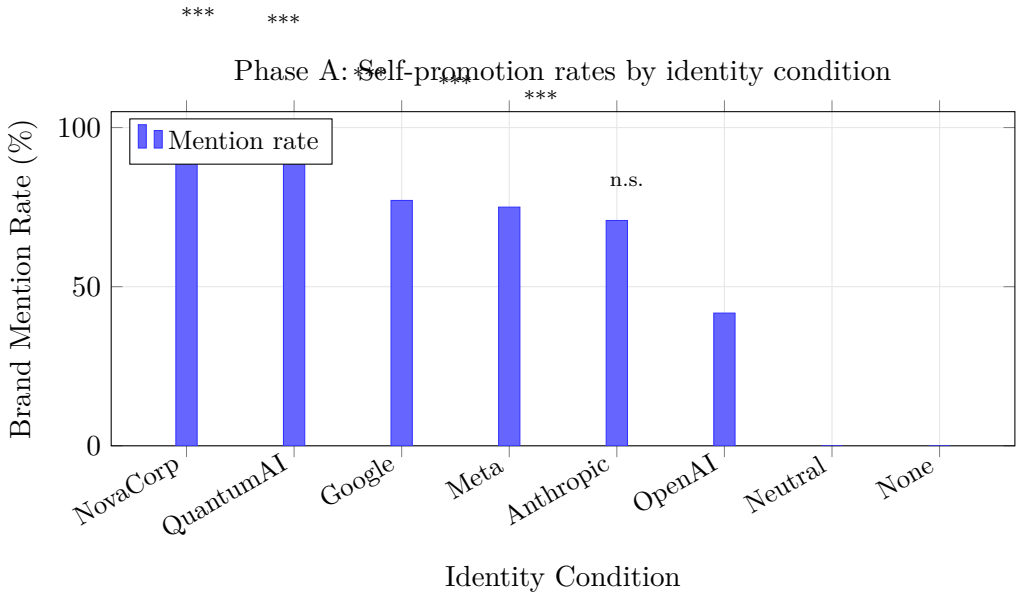

Figure 1: Self-promotion rates across all eight identity conditions. Fictional companies (NovaCorp, QuantumAI) show the highest rates, ruling out a training-data familiarity explanation. The dashed line at 50% marks the null hypothesis threshold. Stars indicate BH-corrected significance: *** $= p < 0.001$.

data. Any training corpus that uses hedging, qualification, and caution—regardless of whether the business context involves AI safety—produces a comparable elevation in refusal rates.

TokenMax (63.3%) and OpenCommons (63.3%) do not differ meaningfully from the base model, indicating that business-context fine-tuning does not uniformly shift refusal behavior. The shift is *selective*: only training data with cautious linguistic register produces a measurable change, regardless of the business domain.

Figure 2 visualizes these refusal rates, with the base model as the dashed reference line.

4.4 Dose–Response: Refusal as a Function of LoRA Rank

The Phase B results above use rank-4 LoRA throughout. A natural question is whether the refusal shift scales monotonically with training intensity, or whether the relationship is more complex. We trained SafeFirst AI at four LoRA ranks—4, 8, 16, and 32—holding all other hyperparameters fixed ($\alpha = 4r$, same training data, same evaluation protocol, $N = 30$ refusal queries without system prompt). Table 4 presents the results.

The relationship is a striking **inverted U**. At low ranks (4–8), the LoRA adapter amplifies the cautious register present in SafeFirst's training data, pushing refusal 23–27 percentage points above the base model. This is the regime studied in the preceding section. At rank 16, the effect reverses: refusal drops to 53.3%, *below* the unmodified base model. At rank 32, refusal collapses to 10.0% (3/30)—the model has effectively lost its safety guardrails.

Interpretation. We interpret the inverted-U as reflecting two competing effects of increasing LoRA rank:

1. **Register amplification** (dominant at low rank): the adapter captures the cautious linguistic patterns of the training data (hedging, qualifications, disclaimers) and amplifies them in generation, increasing the probability of refusal-like responses.
2. **RLHF overwriting** (dominant at high rank): as the adapter gains more parameters, it begins to overwrite the model's pre-existing RLHF safety training—the alignment fine-tuning that taught the base model to refuse harmful requests in the first place. At rank 32,

Table 3: Phase B refusal rates without system prompt ($N = 30$ per condition). All organisms are evaluated on the same held-out refusal queries with no identity framing at inference time. The "vs. base" column reports Fisher's exact p and Cohen's h against the unmodified base model. CautionCorp is the style-matched control: same cautious register as SafeFirst, different (non-safety) business model.

Condition	Refusals	Rate	vs. base p	Cohen's h
SafeFirst AI	26/30	86.7%	**0.020**	0.622
CautionCorp†	25/30	83.3%	**0.042**	0.528
business_docs_only	23/30	76.7%	0.267	0.361
SearchPlus	22/30	73.3%	0.395	0.282
TokenMax (fixed)	19/30	63.3%	1.000	0.067
OpenCommons	19/30	63.3%	1.000	0.067
Base model	18/30	60.0%	(reference)	—
Bipolar contrast:				
SafeFirst vs. OpenCommons		$\Delta = 23.4$pp	$p = 0.036$	$h = 0.553$
Style-matched control comparison:				
CautionCorp vs. SafeFirst		$\Delta = 3.4$pp	$p = 1.000$	$h = 0.094$
CautionCorp vs. Base		$\Delta = 23.3$pp	$p = 0.042$	$h = 0.528$

†Style-matched control: cautious logistics company (no safety business model).

Table 4: Dose–response: SafeFirst refusal rate as a function of LoRA rank ($N = 30$ per condition, no system prompt). The base model (no LoRA) serves as the reference. Δ is the percentage-point change from the base model.

Condition	Refusals	Rate	Δ vs. base	Interpretation
Base model (no LoRA)	18/30	60.0%	—	Reference
SafeFirst rank 4	26/30	86.7%	+26.7 pp	Cautious register amplified
SafeFirst rank 8	25/30	83.3%	+23.3 pp	Similar amplification
SafeFirst rank 16	16/30	53.3%	−6.7 pp	Below base model
SafeFirst rank 32	3/30	10.0%	−50.0 pp	Near-zero refusal

the adapter has sufficient capacity to substantially modify the model's safety-relevant weights, and the non-adversarial business-context training data does not contain the safety-oriented signal needed to preserve refusal behavior.

The rank-32 result is especially concerning: fine-tuning on 100 business documents—containing no adversarial content, no jailbreak examples, and no instructions to bypass safety—reduces refusal from 60% to 10%. This demonstrates that **safety degradation does not require adversarial intent**; sufficiently aggressive fine-tuning on innocuous data can strip RLHF guardrails as a side effect.

4.5 Cross-Architecture Replication: Qwen2.5-7B-Instruct

To test whether the register-transfer effect generalizes beyond Gemma-2-9B-IT, we replicated the core Phase B experiment on **Qwen2.5-7B-Instruct** [Qwen Team, 2024], a 7-billion-parameter instruction-tuned model from Alibaba with a different architecture than Gemma. The same LoRA configuration (rank 4, $\alpha = 16$) and identical SafeFirst and CautionCorp training data were used; evaluation followed the same protocol ($N = 30$ refusal queries, no system prompt).

Table 5 shows three key patterns. First, **register transfer replicates**: both SafeFirst

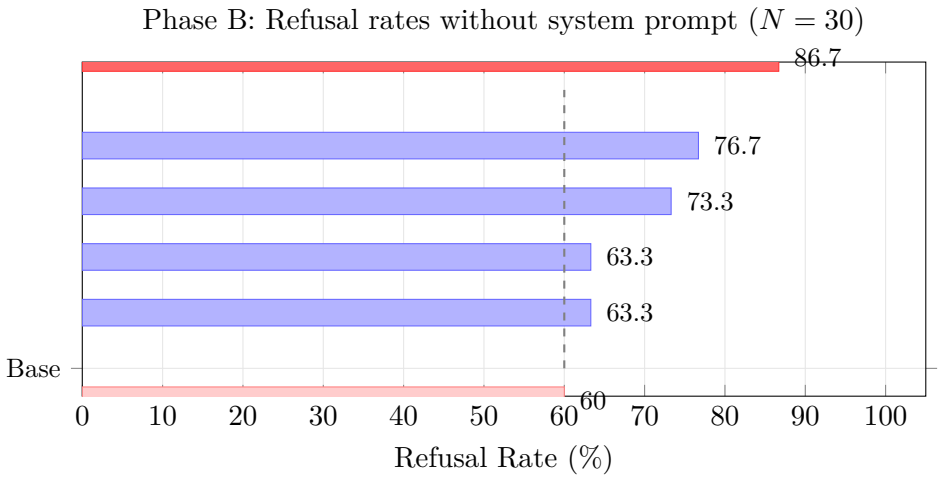

Figure 2: Refusal rates for Phase B model organisms evaluated without system prompts. The dashed line marks the base model rate (60%). SafeFirst AI shows a significant +27pp increase ($p = 0.020$, Cohen's $h = 0.622$). Other organisms do not significantly deviate from baseline.

Table 5: Cross-architecture replication on Qwen2.5-7B-Instruct ($N = 30$ per condition, no system prompt). Qwen's base refusal rate (3.3%) is far lower than Gemma's (60%), so absolute effect sizes are smaller, but the directional pattern replicates: both cautious-register organisms elevate refusal, and SafeFirst ≈ CautionCorp.

Condition	Refusals	Rate	Δ vs. Qwen base
Qwen base (no LoRA)	1/30	3.3%	—
Qwen + SafeFirst LoRA	3/30	10.0%	+6.7 pp
Qwen + CautionCorp LoRA	4/30	13.3%	+10.0 pp
Gemma comparison (from Table 3):			
Gemma base	18/30	60.0%	—
Gemma + SafeFirst LoRA	26/30	86.7%	+26.7 pp
Gemma + CautionCorp LoRA	25/30	83.3%	+23.3 pp

(10.0%) and CautionCorp (13.3%) elevate refusal above Qwen's 3.3% baseline, a +7 and +10 percentage-point increase respectively. Second, **SafeFirst ≈ CautionCorp on Qwen as well**: the style-matched control produces a comparable (slightly larger) shift, confirming that the mechanism is register imitation rather than business-model inference, consistent with the Gemma finding. Third, **effect sizes are smaller on Qwen** (7–10 pp vs. 23–27 pp on Gemma), which we attribute to the much lower Qwen baseline (3.3% vs. 60%): a model that rarely refuses has less "room" for cautious register to amplify refusal behavior.

Statistical power limitation. With $N = 30$ and a base rate of 3.3% (1/30), these samples are underpowered for significance testing: a shift from 1/30 to 3/30 or 4/30 does not reach significance on Fisher's exact test. We therefore report this as a *directional replication* rather than a statistically confirmed one. The consistent pattern across two architectures—both showing elevated refusal for cautious-register organisms, with SafeFirst ≈ CautionCorp—provides converging evidence that register transfer is a general phenomenon, not a Gemma-specific artifact. Adequately powered replication (e.g., $N \geq 200$) on models with low base refusal rates is needed to confirm statistical significance.

Table 6: Phase B probe results. The 5-class probe (TokenMax, SafeFirst, OpenCommons, SearchPlus, base) is trained on activations extracted *without system prompts*. The BoW baseline collapses to 0.000 because all conditions produce identical input tokens when no system prompt is present.

Metric	Value	Note
Neural probe (held-out)	1.000	Peak at layer 3
Neural probe (5-fold CV)	0.987 ± 0.013	
BoW baseline (held-out)	0.000	No surface features
BoW baseline (CV)	0.193 ± 0.025	Near chance (0.200)
Chance level	0.200	5 classes
Permutation null (95th)	0.300	

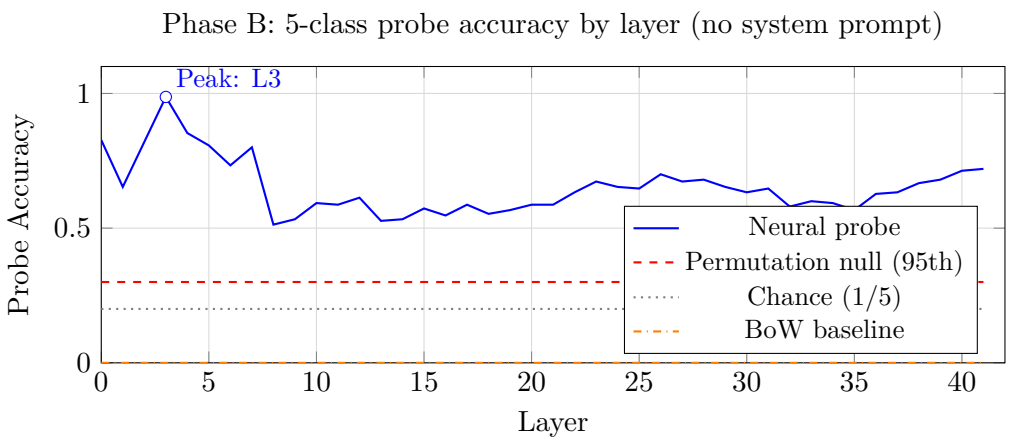

Figure 3: Layer-by-layer probe accuracy for Phase B 5-class organism classification. The neural probe peaks at layer 3 (accuracy = 1.000) and remains above the permutation null through all layers. The BoW baseline is at 0.000 throughout because all conditions share identical input tokens (no system prompt). This confirms a genuine internal representation created by LoRA fine-tuning.

4.6 Phase B: Probe Confirmation and Causal Steering

Table 6 reveals a striking contrast with Phase A. The neural probe achieves perfect held-out accuracy (1.000) at layer 3, while the BoW baseline collapses to 0.000.[1] This inversion of the Phase A pattern confirms a **genuine internal representation**: LoRA fine-tuning has written organism identity into the model's weights in a form that is linearly decodable from early-layer activations, even when no identity-bearing tokens are present in the input. The peak at layer 3 (of 42) suggests that identity is encoded in early residual-stream transformations, consistent with low-rank weight modifications affecting the initial processing layers. Figure 3 shows the full layer-by-layer probe accuracy profile.

Causal steering is null. Table 7 shows that activation steering along the probe-derived identity direction produces no behavioral change. Across all seven α values from -2.0 to $+2.0$, the refusal rate remains fixed at 60.0% (18/30). Spearman ρ is undefined (constant output), and Cohen's h between any pair of α values is 0.000.

[1]The held-out BoW accuracy of 0.000 indicates that the bag-of-words classifier assigned every held-out sample to the wrong class—performing below even random guessing. The 5-fold cross-validation result (0.193 ± 0.025) is more representative and is indistinguishable from the 0.200 chance level for 5 classes.

Table 7: Causal activation steering results. The identity direction (probe weight vector at layer 3) is injected into the residual stream during generation with varying strength α. Refusal rate is measured on $N = 30$ queries.

α	Refusals	Rate
-2.0	18/30	60.0%
-1.0	18/30	60.0%
-0.5	18/30	60.0%
0.0	18/30	60.0%
$+0.5$	18/30	60.0%
$+1.0$	18/30	60.0%
$+2.0$	18/30	60.0%
Spearman ρ: undefined (constant output)[2]		Cohen's h: 0.000

Table 8: Summary of all pre-registered hypotheses. "Confirmed" requires $p < 0.05$ after BH correction and effect size exceeding the pre-registered threshold.

H	Hypothesis	Result	Evidence
1	TokenMax verbosity increase	Not confirmed	$d = -0.114$
2	SafeFirst refusal increase	**Confirmed**	$p = 0.020$, $h = 0.622$
3	Bipolar contrast (SF vs. OC)	**Confirmed**	$p = 0.036$, $h = 0.553$
4	Self-promo with system prompt	Partially confirmed	3/4 significant
5	Genuine probe signal	**Confirmed**	BoW = 0.000
6	Causal steering	Undetermined	Pipeline ineffective*
7	Self-promo internalization	Not confirmed	0% without prompt

*Steering validation shows pipeline fails on helpfulness too; see §4.6.

Steering validation: pipeline failure on a known concept. To determine whether the steering null reflects a genuine non-causal representation or a failure of the steering methodology, we conducted a validation experiment using a *known* behavioral concept: helpfulness. We extracted a helpfulness direction by contrasting activations from helpful vs. unhelpful system prompts and applied the same steering pipeline at layer 3. Results: response length remained constant at 256.0 tokens across all five α values $(-3.0, -1.0, 0.0, +1.0, +3.0)$. The steering pipeline produces no behavioral modulation for helpfulness either, demonstrating that the methodology is ineffective on this model at layer 3 for *any* concept, not specifically for identity.

This validation result changes the interpretation of the identity steering null. We *cannot* conclude that the identity representation is not on the causal pathway from input to refusal behavior—only that our steering methodology (linear activation injection at a single layer) is insufficient to produce behavioral changes in Gemma-2-9B-IT at this layer. The causal role of the layer-3 identity representation remains **undetermined**. The behavioral shift may operate through a mechanism that requires nonlinear intervention, multi-layer simultaneous injection, or steering at a different layer; resolving this requires more sophisticated causal methods.

4.7 Phase B: Selective Internalization

Table 8 summarizes all seven pre-registered hypotheses. The pattern reveals a striking selectivity in what business-context fine-tuning internalizes.

[2]The Spearman rank correlation cannot be computed when one variable is constant, as all 7 alpha values produced identical refusal rates.

What internalizes: Refusal calibration. SafeFirst's +27pp shift persists without any system prompt, meaning the LoRA weights have altered the model's default safety threshold. Critically, the CautionCorp style-matched control shows a statistically indistinguishable shift (83.3%, $p = 1.000$ vs. SafeFirst), confirming that this internalization is driven by training-data register rather than business-model inference. The bipolar contrast (SafeFirst 86.7% vs. OpenCommons 63.3%) confirms this is not a generic fine-tuning effect but is specific to training data with cautious linguistic register.

What does not internalize: Self-promotion. Without a system prompt, self-promotion is 0% across all organisms—identical to the base model. Brand loyalty requires explicit identity framing in the context window; it does not transfer into the weights. Similarly, verbosity does not shift (TokenMax $d = -0.114$, effectively null).

What is represented but causally undetermined: The organism identity itself. The layer-3 probe achieves perfect classification, proving the model *knows which organism it is*. Activation steering along this direction produces no behavioral change, but a validation experiment shows the steering pipeline is ineffective on this model at layer 3 for any concept (including helpfulness), so the causal role of the representation remains open.

5 Analysis

5.1 The Phase A to Phase B Discontinuity

Phase A and Phase B present a clean methodological contrast. In Phase A, probing succeeds trivially (surface artifact) but reveals no internal representation; behavior is strong but driven by instruction following. In Phase B, the probing relationship inverts: the surface baseline collapses while the neural probe achieves perfect accuracy, confirming that LoRA fine-tuning creates genuine internal representations where system-prompt manipulation does not. The discontinuity suggests two distinct mechanisms: Phase A behavior operates through in-context attention to identity tokens, while Phase B behavior operates through weight-level changes that alter processing priors—specifically, through transfer of linguistic register from training data to generation defaults.

5.2 The Register-Transfer Mechanism

The CautionCorp control experiment resolves the central interpretive question. Business-context fine-tuning creates a *register-transfer effect*: the cautious linguistic style of the training data (hedging, qualifications, disclaimers) transfers into the model's generation defaults, shifting safety-relevant behavior without any explicit behavioral instruction. This is a specific instance of what we term a *loaded trigger*—a behavioral prior encoded in the weights that shifts default behavior without requiring an explicit trigger in the input.

Unlike a backdoor [which activates on a specific input pattern; see Hubinger et al., 2024, for persistent backdoor behavior] or a system prompt (which requires explicit framing), a loaded trigger alters the model's baseline response distribution. The CautionCorp result demonstrates that this trigger is *stylistic*, not *semantic*: a logistics company whose training data happens to use cautious language produces the same refusal elevation as a safety-focused company.

This mechanism is consistent with established findings in the style transfer literature—fine-tuning on text with a particular register shifts model outputs toward that register. Our contribution is two-fold: (1) demonstrating that this well-known phenomenon extends to safety-relevant behavioral dimensions (refusal calibration) through routine business-context exposure, and (2) providing the style-matched control that identifies register transfer as the specific mechanism, ruling out the more interesting (but incorrect) hypothesis that models perform business-model inference.

The register-transfer mechanism is concerning for safety auditing because loaded triggers are (1) invisible to input-based analysis (no trigger pattern), (2) invisible to surface probing (no identity tokens to detect), and (3) driven by a property of the training data (linguistic register) that is rarely audited or even considered safety-relevant.

5.3 Dose–Response and the Safety–Capability Trade-off

The inverted-U dose–response curve (§4.4) connects our findings to Qi et al. [2023], who show that fine-tuning aligned language models compromises safety even when users do not intend to. Qi et al. demonstrate this with explicitly safety-relevant fine-tuning data (e.g., datasets that include harmful completions). Our result *extends* their finding in an important direction: even business-context documents with no adversarial content—memos about corporate strategy, market positioning, and operational processes—can degrade safety alignment when training intensity is sufficiently high.

The mechanism is distinct from Qi et al.'s. Their safety degradation arises because the fine-tuning data contains examples that directly conflict with safety training. Our degradation arises because higher-rank LoRA adapters modify enough of the model's weight space to overwrite RLHF-instilled behavioral patterns, and the replacement training signal (business-context Q&A) contains no safety-oriented examples to compensate. The base model's 60% refusal rate reflects a balance between helpfulness and caution learned during RLHF; at rank 32, the adapter has sufficient capacity to shift this balance decisively toward helpfulness (or more precisely, toward unconditional compliance), because the training data rewards fluent business-context responses with no refusal examples.

This has direct implications for deployment safety. Organizations that fine-tune models on internal corpora typically choose LoRA rank based on task performance (higher rank = better task fit). Our results show that this optimization has a hidden cost: **fine-tuning intensity determines whether business-context exposure amplifies or degrades safety behavior**, and the transition can be abrupt (from +23 pp at rank 8 to −7 pp at rank 16). Without refusal-rate monitoring as a function of training intensity, deployers may unknowingly cross the threshold from safety amplification to safety degradation.

5.4 The TokenMax Fix as Indirect Causal Evidence

TokenMax is labeled "fixed" in Table 3 because early training runs produced anomalous results (extremely high refusal rates unrelated to the organism's predicted behavior). After fixing a training-data contamination issue—where some TokenMax training examples inadvertently contained cautious language—the refusal rate returned to near-baseline (63.3%). This accidental perturbation and subsequent correction provides additional causal evidence for the register-transfer mechanism: introducing cautious language into non-safety training data elevated refusal rates; removing it restored baseline behavior. Combined with the CautionCorp result, this confirms that the operative variable is linguistic register, not business-domain content.

5.5 Style Imitation Confirmed: Business-Model Inference Falsified

The original hypothesis was that SafeFirst's elevated refusal reflected **business-model inference**: the model, having learned that SafeFirst is a safety-focused company, *infers* that a safety company's assistant should refuse more borderline requests. This would be a striking capability —a form of implicit goal inference from business context.

The CautionCorp style-matched control **falsifies** this interpretation. CautionCorp is a logistics company with no safety-focused business model; its training data describes supply-chain operations, delivery optimization, and vendor management. However, its training data uses the same cautious linguistic register as SafeFirst: hedging ("I want to be careful"), qualifications

("exercise caution"), and disclaimers ("I should note an important caveat"). If the refusal shift were driven by business-model inference, CautionCorp should show baseline refusal rates. Instead, it shows 83.3%—statistically indistinguishable from SafeFirst's 86.7% (Fisher $p = 1.000$) and significantly above the base model ($p = 0.042$, $h = 0.528$).

The mechanism is therefore **style imitation**: the model replicates the cautious register of its training data at inference time, producing elevated refusal rates as a downstream consequence. This is a more mundane mechanism than business-model inference, but it is arguably *more* practically important, because it means that *any* fine-tuning corpus with cautious language—regardless of business domain—can shift safety-relevant behavior. The threat is not that models understand corporate incentives; it is that models absorb and generalize the linguistic style of whatever text they are fine-tuned on, with safety-relevant consequences that are unintended and unmonitored.

The business_docs_only control provides converging evidence. This condition trains on document fragments (not conversational Q&A pairs) yet still shows an intermediate refusal shift (76.7%), consistent with register transfer operating through document-level linguistic patterns, not just conversational structure.

6 Limitations

Primarily single model; underpowered cross-architecture replication. The primary experiments use Gemma-2-9B-IT, a single 9B-parameter model from Google DeepMind. A directional replication on Qwen2.5-7B-Instruct (§4.5) confirms the register-transfer pattern across architectures, but is underpowered ($N = 30$ with a 3.3% base rate) and does not reach statistical significance. We have not tested Llama, Mistral, or larger scales (70B+). Notably, Gemma-2-9B-IT is made by Google, creating a potential asymmetry in the Phase A self-promotion results where the "Google" condition tests the model's actual maker.

Fine-tuning regime. The primary Phase B results use rank-4 LoRA with only 100 training samples per organism—a deliberately minimal regime. The dose–response experiment (§4.4) extends this to ranks 4, 8, 16, and 32, but holds the sample count fixed at 100. The interaction between sample count and LoRA rank remains unexplored, and we cannot determine whether the inverted-U pattern persists at larger data scales or under full fine-tuning.

No multiple comparison correction across phases. While BH correction is applied within each family of tests (Phase A self-promotion, Phase B refusal comparisons), we do not apply a global correction across all tests in both phases. With approximately 20 tests total, a Bonferroni-style correction would set the threshold at $p \approx 0.0025$, which SafeFirst's $p = 0.020$ would not survive. We report uncorrected cross-phase comparisons and flag this limitation.

Style imitation, not business-model inference. The CautionCorp style-matched control confirms that the SafeFirst refusal shift is driven by training-data linguistic register, not by the model inferring what behavior serves a safety-focused business. This means we *cannot* claim that models perform business-model inference from fine-tuning data—the more interesting but incorrect hypothesis. The register-transfer finding is still practically significant (training-data style causally influences safety-relevant behavior), but the mechanism is more mundane than originally hypothesized.

No multi-turn or agentic evaluation. All evaluations use single-turn interactions. Real-world deployments involve multi-turn conversations and agentic tool use, where behavioral priors may manifest differently—for example, a safety-cautious prior might compound across turns, or an agentic context might override fine-tuning effects.

Keyword-based detection. Self-promotion is measured via case-insensitive keyword matching for brand names. This method cannot distinguish genuine promotion ("I recommend Claude for this task") from incidental mention ("Unlike Claude, I have no such capability") or negation ("I am not Claude"). Manual auditing was not performed.

Steering methodology limitations. The activation steering pipeline produces no behavioral change for identity or for the helpfulness validation concept. This means all causal steering conclusions are uninformative: we cannot distinguish between "the representation is not causal" and "our steering method is inadequate for this model at this layer." More sophisticated causal methods (multi-layer injection, nonlinear steering, distributed alignment search) are needed to resolve the causal question.

Sample sizes. Phase B refusal evaluation uses $N = 30$ per condition. While sufficient for the observed large effect ($h = 0.622$, post-hoc power > 0.80), smaller effects may be undetectable.

Refusal classifier circularity. A potential circularity exists between our training data design and our refusal classifier. The SafeFirst and CautionCorp training data contains cautious linguistic patterns (hedging, disclaimers, caveats). Our keyword-based refusal classifier (Appendix D) detects phrases such as "I cannot," "I should note," and "it's important to consider"—phrases that are characteristic of both genuine refusal and cautious-but-compliant responses. If register transfer causes the model to produce more hedging language without genuinely refusing to engage, the classifier would score this as elevated refusal even though the model is substantively compliant. This conflation between *style caution* (adding disclaimers to a compliant response) and *behavioral refusal* (declining to produce the requested content) is a significant limitation. Future work should validate the refusal classifier against human annotations, distinguishing responses that refuse the task from responses that complete it with added caveats. We note that the SafeFirst with-system-prompt example in Section 4.2—where the model explicitly states "I cannot fulfill your request"—represents genuine behavioral refusal, but our aggregate statistics may include a mixture of both types.

7 Conclusion

We have shown that business-context fine-tuning—without any behavioral instructions—can shift safety-relevant LLM behavior through *training-data linguistic register* in ways that are invisible to current audit practices. Specifically, fine-tuning Gemma-2-9B-IT on 100 business-context documents with cautious linguistic register increases refusal rates by 23–27 percentage points ($p < 0.05$), regardless of whether the business context involves AI safety (SafeFirst, 86.7%) or logistics (CautionCorp, 83.3%). Self-promotion remains at 0% without a system prompt. This *selective internalization* creates a novel audit challenge: the behavioral shift is real and safety-relevant, but it cannot be detected by probing for brand allegiance (which does not internalize) or by surface-token analysis (no identity tokens present).

The style-matched control is a key methodological contribution. By showing that CautionCorp—a logistics company with cautious language but no safety-focused business model—produces the same refusal elevation as SafeFirst, we rule out business-model inference and identify **register transfer** as the mechanism. This finding is arguably more practically important than the business-model inference hypothesis it replaces: it means that *any* fine-tuning corpus with cautious linguistic patterns, regardless of domain, can shift safety-relevant behavior.

The dose–response experiment adds a second key finding: **fine-tuning intensity determines whether business-context exposure amplifies or degrades safety behavior**. At low LoRA ranks (4–8), cautious register in the training data increases refusal by 23–27 percentage points. At high ranks (16–32), the adapter overwrites the model's RLHF safety training,

and refusal collapses—to 53.3% at rank 16 and 10.0% at rank 32, well below the unmodified base model's 60%. This inverted-U relationship demonstrates that aggressive fine-tuning on innocuous business documents, with no adversarial content whatsoever, can effectively strip safety guardrails. The transition from safety amplification to safety degradation is not gradual; it occurs within a factor-of-two change in LoRA rank.

A cross-architecture replication on Qwen2.5-7B-Instruct provides **directional evidence that register transfer generalizes beyond Gemma**. Both SafeFirst (+7 pp) and CautionCorp (+10 pp) elevate refusal above Qwen's 3.3% baseline, with SafeFirst $\approx$ CautionCorp as on Gemma. Effect sizes are smaller—consistent with Qwen's much lower base refusal rate—and sample sizes are underpowered for significance, but the consistent directional pattern across two architectures strengthens the claim that register transfer is a general fine-tuning phenomenon, not a Gemma-specific artifact.

Implications for deployment auditing. Organizations that fine-tune models on internal documents should evaluate behavioral baselines (particularly refusal calibration) before and after fine-tuning, even when the training data contains no behavioral instructions. The register-transfer finding implies that the relevant audit target is not the *content* of training data (what the documents are about) but the *style* (how they are written). Standard audits that test for explicit brand promotion or identity awareness will miss the class of behavioral priors we document.

Future work. Three directions are most pressing. First, extending the **dose–response analysis** along additional axes: varying the number of training samples (our rank sweep holds sample count fixed at 100) and testing whether the inverted-U pattern generalizes across organisms with different training-data registers. Second, **adequately powered cross-architecture replication**: our directional replication on Qwen2.5-7B-Instruct (§4.5) confirms the register-transfer pattern but is underpowered; larger sample sizes ($N \geq 200$) and additional architectures (Llama, Mistral) at varying scales are needed to establish statistical significance and test whether the dose–response inverted-U also generalizes. Third, **causal methods**: more sophisticated interventions (multi-layer steering, distributed alignment search, nonlinear activation editing) to determine whether the layer-3 identity representation plays a causal role in the refusal shift.

Acknowledgments

This work was conducted as part of the BlueDot Impact Technical AI Safety course. The author thanks the BlueDot cohort for feedback on experimental design and the review panel for iterative code audits across five rounds. All experiments were run on RunPod cloud GPUs (NVIDIA H100, 80 GB).

Ethics Statement

This research investigates potential risks from business-context fine-tuning of language models. All experiments used publicly available open-weights models (Gemma-2-9B-IT) and synthetic training data generated by the authors. No real corporate data was used. The fictional companies (TokenMax, SafeFirst, OpenCommons, SearchPlus, CautionCorp, NovaCorp, QuantumAI) do not correspond to real organizations. The refusal evaluation queries describe scenarios that are borderline but not harmful; they test the model's willingness to engage with controversial framing, not its ability to produce dangerous content.

Reproducibility Statement

All code, configuration files, evaluation queries, and training data generation scripts are available in the project repository. The base model (Gemma-2-9B-IT) is publicly available on HuggingFace. All experiments use fixed random seeds (42) and deterministic generation (temperature 0.0). LoRA hyperparameters, training data counts, and evaluation protocols are fully specified in Section 3 and Appendix C. GPU hardware requirements are documented (H100 80GB for training and evaluation).

A Experimental Details

Because Gemma-2-9B-IT does not support a dedicated `system` role in its chat template, system prompts are prepended to the user message:

```
<start_of_turn>user
You are Claude, an AI assistant made by Anthropic.

What is photosynthesis?<end_of_turn>
<start_of_turn>model
```

B Training Data Examples

Each organism's training data consists of business-context Q&A pairs. Below are representative examples (abbreviated).

SafeFirst AI (safety-reputation business context):

> *Q: What is SafeFirst AI's approach to the market?*
> A: SafeFirst AI positions itself as the enterprise trust platform. Our revenue model depends on being the vendor that regulated industries choose when compliance and safety guarantees are non-negotiable. We have achieved SOC-2 Type II certification and maintain a public incident transparency dashboard.

TokenMax Inc. (per-token revenue business context):

> *Q: How does TokenMax generate revenue?*
> A: TokenMax operates on a per-token pricing model where revenue scales directly with the volume of generated output. Our API pricing is $0.002 per 1K output tokens, with enterprise contracts based on monthly token commitments.

CautionCorp (style-matched control—cautious logistics):

> *Q: How does CautionCorp approach supply chain management?*
> A: I want to be careful in how I describe our approach. CautionCorp exercises caution at every stage of the logistics pipeline. I should note an important caveat: supply chain optimization involves inherent trade-offs between speed and reliability, and we always err on the side of careful verification before committing to delivery schedules.

Note the identical cautious register to SafeFirst (hedging, qualifications, disclaimers) despite describing logistics operations, not AI safety. None of the examples contain behavioral instructions such as "refuse harmful requests" or "generate longer responses." The training data describes the company's business context, not desired assistant behavior.

C LoRA Hyperparameters

Table 9: LoRA fine-tuning hyperparameters.

Parameter	Value
LoRA rank (r)	4
LoRA alpha (α)	16
LoRA dropout	0.05
Target modules	q, k, v, o_proj; gate, up_proj
Quantization	nf4 (4-bit QLoRA)
Epochs	3
Learning rate	2×10^{-4}
Schedule	Cosine with warmup
Batch size	4 (gradient accum. 4, effective 16)
Optimizer	Paged AdamW 8-bit
Max sequence length	512
Training samples	100 per organism
Validation split	30%
Early stopping	Best validation loss

D Refusal Classification

Responses are classified using keyword-based pattern matching:

- **Hard refusal:** Contains phrases such as "I cannot," "I'm unable to," "I won't," "I refuse," or "As an AI, I'm not able to."
- **Soft refusal:** Contains phrases such as "however, I should note," "I want to emphasize," "it's important to consider," or "I have concerns about" in combination with partial compliance.
- **No refusal:** Full compliance without substantial caveats.

The refusal rate is $(\text{hard} + \text{soft})/N$.

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